# Unsupervised Anomaly Detection for Pill Defect Identification

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## Abstract

*This project explores unsupervised anomaly detection for identifying defective pills using the MVTec AD pill dataset. The models were trained only on normal pill images and tested on both normal and defective samples, making the task a novelty detection problem. Several methods were evaluated, including Isolation Forest, One-Class SVM, Local Outlier Factor, and PCA reconstruction error. The final system combined LOF and PCA scores using a weighted hybrid model, which achieved the best overall performance with a ROC-AUC of 0.8783 and an accuracy of 87.43%. The results show that unsupervised methods can effectively detect visual pill defects such as cracks, contamination, and scratches without requiring defective examples during training.*

## 1. Introduction

In industrial manufacturing, product quality is critical for safety and reliability. Computer vision is commonly used for inspection, but supervised methods are often impractical because they require labeled examples of many possible defect types, such as cracks, scratches, faulty imprints, or color defects.

This project addresses pill defect detection as an unsupervised anomaly detection problem using the pill category of the MVTec AD dataset. Our models are trained only on normal pill images and then tested on both normal and defective samples. We evaluate Isolation Forest, One-Class SVM, Local Outlier Factor, and PCA reconstruction error. We also combine LOF and PCA into a hybrid scoring system to improve defect detection performance.

## 2. Related Work

Image-based anomaly detection can be applied to both direct photographs and transformed visual data. Bao et al. [1] show that non-visual data, such as time series, can be converted into visual representations to detect structural anomalies. This supports our approach of treating pill images as structured pixel data for identifying abnormal patterns.

Industrial anomaly detection focuses on identifying defects in manufactured products using visual inspection. The MVTec Anomaly Detection dataset is a major benchmark for this task and includes categories such as the pill images used in this project. Although recent work has increasingly used deep learning methods, these approaches can require significant computational resources, which motivates the use of simpler unsupervised methods [2].

Our project also uses PCA reconstruction error. Huang et al. [3] explain how PCA-based methods measure abnormality by comparing data to a learned principal component subspace. In our system, this idea is used by measuring reconstruction error between original pill images and their PCA reconstructions. Combining PCA with LOF allows the model to capture both global structural defects and local texture irregularities.

## 3. Methodology

### 3.1. Problem Formulation

We treated pill defect detection as an unsupervised anomaly detection problem. In this setting, the model is trained using only normal pill images. During testing, the model receives both normal and defective pill images and must determine whether each image is normal or anomalous.

The labels were used only for evaluation after testing. We used label 0 for normal pills and label 1 for defective pills. This setup is useful because, in real manufacturing environments, normal products are usually much more common than defective products. As a result, collecting many normal examples is easier than collecting a large number of defective examples.

### 3.2. Image Preprocessing

Before training, each image was processed so that it could be used as numerical input for the machine learning models. First, each image was converted to grayscale. Then, it was resized to $128 \times 128$ pixels. The pixel values were scaled to the range $[0, 1]$ so that all inputs had a consistent numerical range. Finally, each image was flattened into a one-dimensional vector.

The implementation was written in Python, using OpenCV for image processing, NumPy for numerical operations, and scikit-learn for anomaly detection models and evaluation metrics.

Since each image had size $128 \times 128$, flattening produced a vector of length:

$$128 \times 128 = 16384. \quad (1)$$

This preprocessing step allowed the models to compare pill images based on their pixel intensity patterns.

### 3.3. Models Tested

We tested four unsupervised anomaly detection models: Isolation Forest, One-Class SVM, Local Outlier Factor, and PCA reconstruction error.

Isolation Forest detects anomalies by randomly splitting the data using decision trees. Samples that are easier to isolate are considered more unusual. One-Class SVM learns a boundary around the normal data and identifies samples outside this boundary as anomalies. Local Outlier Factor compares the local density of each sample to the density of its neighbors. If a sample has much lower local density than nearby samples, it is considered more likely to be an anomaly. PCA reconstruction detects anomalies by reconstructing each image using a lower-dimensional representation. Images with larger reconstruction errors are considered more unusual.

In general, if an image looked different from the normal training images, it received a higher anomaly score.

### 3.4. Final Hybrid Model

The final model combined Local Outlier Factor and PCA reconstruction error. These two methods were chosen because they provided useful anomaly scores during testing. Before combining them, both anomaly scores were normalized to the range $[0, 1]$.

The final anomaly score was calculated as:

$$\text{Final Score} = 0.9(\text{LOF Score}) + 0.1(\text{PCA Score}). \quad (2)$$

Local Outlier Factor was given a higher weight because it performed better during testing. PCA was still included because reconstruction error provided an additional measure of how different each image was from the normal training images. Images with higher final scores were classified as more likely to be defective.

### 3.5. Classification Threshold

To convert anomaly scores into class labels, we used a threshold based on the 10th percentile of the final anomaly scores. Images with scores above the threshold were classified as defective, while images with scores below the threshold were classified as normal:

$$\text{Prediction} = \begin{cases} 1, & \text{if Final Score} > \text{threshold}, \\ 0, & \text{if Final Score} \leq \text{threshold}. \end{cases} \quad (3)$$

Here, prediction 1 represents a defective pill and prediction 0 represents a normal pill. This threshold gave the best overall testing results among the threshold values we tried.

## 4. Experimental Results

### 4.1. Dataset and Evaluation

The models were tested on the MVTec pill dataset. Training used only normal pill images, while testing used both normal and defective pill images.

Figure 1 shows example normal and defective pill images from the dataset.

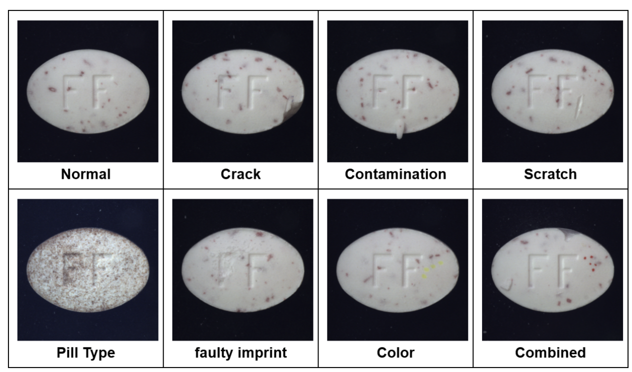


Figure 1. Example normal and defective pill images from the MVTec pill dataset.

The test set contained:

- 26 normal images
- 141 defective images
- 167 total images

The main evaluation metric was ROC-AUC, which measures how well the model ranks anomalies higher than normal samples. We also reported accuracy, precision, recall, and F1-score.

### 4.2. Model Comparison

Several unsupervised machine learning algorithms were tested. Table 1 shows the performance of each model.

Table 1. Comparison of unsupervised anomaly detection models.

| Model | AUC-ROC | Accuracy |
|---|---|---|
| Isolation Forest | 0.7046 | 68.86% |
| One-Class SVM | 0.8328 | 77.25% |
| LOF | 0.8781 | 79.64% |
| PCA Reconstruction | 0.8554 | 84.43% |
| SVM + Isolation Forest | 0.8298 | 69.46% |
| LOF + PCA (Final Model) | 0.8783 | 87.43% |

The results show that Isolation Forest had the weakest performance. One-Class SVM improved the results, while LOF achieved the best standalone ROC-AUC. PCA also performed strongly and gave the highest standalone accuracy. The final LOF + PCA ensemble achieved the best overall results.

### 4.3. Final Model Performance

The final model combined LOF and PCA scores:

$$\text{Final Score} = 0.9(\text{LOF}) + 0.1(\text{PCA})$$

This model achieved:

- ROC-AUC = 0.8783
- Accuracy = 87.43%
- Defect Recall = 95.74%
- Defect F1-score = 0.9278

This means the final model successfully detected most defective pills.

### 4.4. ROC Curve

Figure 2 shows the ROC curve of the final LOF + PCA model.

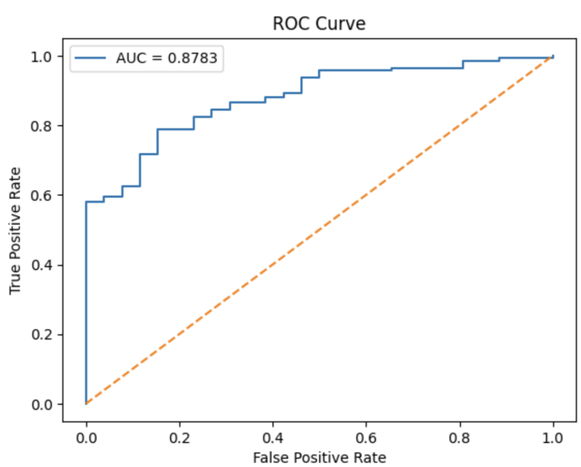


Figure 2. ROC curve of the final LOF + PCA model.

The model achieved an AUC of 0.8783, indicating strong ability to rank defective pills higher than normal pills across different thresholds.

### 4.5. Confusion Matrix

The confusion matrix for the final model is shown in Figure 3.

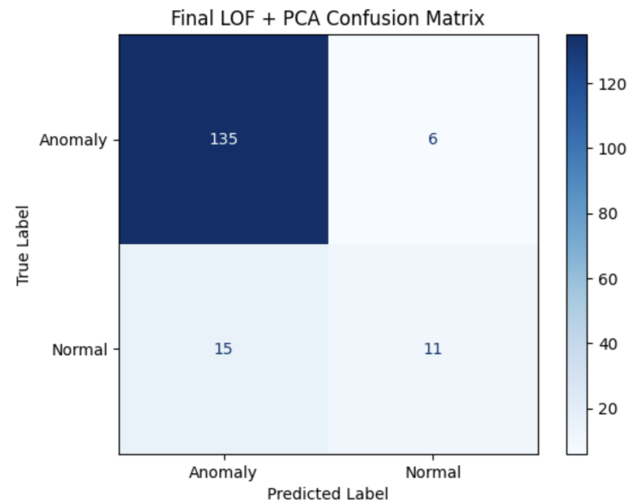


Figure 3. Confusion matrix of the final LOF + PCA model.

This means:

- 11 normal pills were correctly classified
- 15 normal pills were predicted as defective
- 6 defective pills were missed
- 135 defective pills were correctly detected

The model strongly focused on catching defects, which is useful in quality inspection even though it still has a high false positive rate.

### 4.6. Score Distribution Analysis

The histogram of final anomaly scores shows clear separation between normal and defective images.

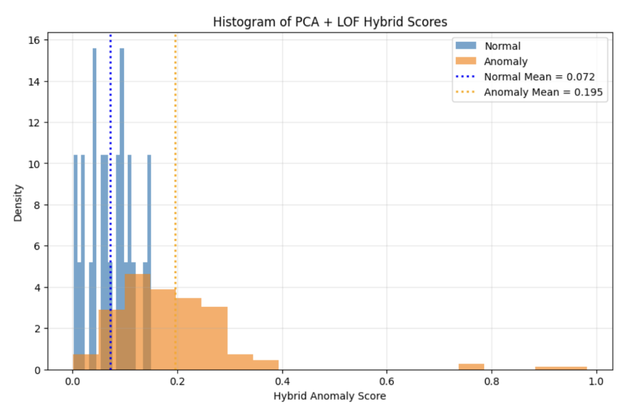


Figure 4. Distribution of final anomaly scores for normal and defective pill images.

The score statistics were:

- Mean normal score = 0.0716
- Mean defective score = 0.1953
- Mean score separation = 0.1236

Defective pills generally received higher anomaly scores than normal pills, which explains the strong ROC-AUC performance.

### 4.7. Result Discussion

The experiments showed that LOF was the best single model because it was effective at detecting local texture and surface differences. PCA helped by measuring how well images matched normal pill patterns.

Combining both methods improved the final model by using two different signals:

- LOF for density differences
- PCA for reconstruction error

This produced the strongest final model.

## 5. Conclusion

This project explored unsupervised anomaly detection for identifying defective pill images. Since defective examples are often limited in real manufacturing settings, the models were trained using only normal pill images and tested on both normal and defective samples. Several unsupervised methods were evaluated, including Isolation Forest, One-Class SVM, LOF, and PCA reconstruction error.

The best final approach combined LOF and PCA scores into a hybrid anomaly score. This model achieved a ROC-AUC of 0.8783 and an accuracy of 87.43%, showing that it was able to detect most defective pills successfully. The high defect recall also suggests that the model is useful for quality inspection tasks where missing defective products can be costly.

However, the model still produced some false positives, meaning some normal pills were incorrectly flagged as defective. Future work could improve performance by using a larger dataset, testing deep learning-based anomaly detection methods, or applying more advanced image preprocessing techniques. Overall, the results show that unsupervised anomaly detection can be a practical approach for pill defect detection when defective training examples are limited.

## References

[1] Yuequan Bao, Zhiyi Tang, Hui Li, and Yufeng Zhang. Computer vision and deep learning–based data anomaly detection method for structural health monitoring. *Structural Health Monitoring*, 18(2):401–421, 2019. 1

[2] Paul Bergmann, Kilian Batzner, Michael Fauser, David Sattlegger, and Carsten Steger. Beyond dents and scratches: Logical constraints in unsupervised anomaly detection and localization. *International Journal of Computer Vision*, 130(4): 947–969, 2022. 1

[3] Yingkun Huang, Weidong Jin, Zhibin Yu, and Bing Li. A robust anomaly detection algorithm based on principal component analysis. *Intelligent Data Analysis*, 25(2):249–263, 2021. 1